

IEDA 1250 Optimizing Decisions for Personal and Business Development

Lecture 1: Introduction

Instructor: Bo YANG

Summer, 2026



Logistics

- Instructor: Bo Yang (yangb@ust.hk)
- Time & Classroom
 - Monday, Wednesday, Friday, 9:00 – 12:20, Room 2464
- Tutorial
 - Monday, Wednesday, Friday, 12:30 – 13:10, Room 2464
- TAs
 - Ms. Hanyan DUAN (hduanag@connect.ust.hk)
 - Mr. Cho Yin NG (cyngba@connect.ust.hk)
 - Mr. Hongfan WU (hwucn@connect.ust.hk)
- Canvas: Announcements about class, lecture notes, HWs, and solutions

Grading

- Two Assignments (20%):
 - Late submission will NOT be accepted, as the solution will be posted on Canvas after the due time.
- Midterm Exam (40%):
 - After Game Theory
- Final Exam (40%):
 - Last class
- **Class attendance: 0%, but helpful for border case**
- Academic Integrity
 - Discussion is allowed for HWs
 - Plagiarism and cheating will be reported to the university

Reference Books

- Introduction to Operations Research, by Hillier, Frederick S., and Gerald J. Lieberman, McGraw-Hill, 2015.
- Operations and Supply Chain Management, by F. R. Jacobs and R. Chase (13th edition), McGraw-Hill, 2018
- Convex Optimization, by Boyd, Stephen, 2004

Today's Outline

- Introduction to operations research (OR)
- Main sub-disciplines of OR
- Main optimization techniques covered by this course
- Brief history of OR

Operations Research

- Operations Research (OR) is a branch of mathematics that deals with the development and application of analytical methods to improve decision making
 - Build mathematical models to “describe” real problems
 - The model contains **decision variables**, **constraints**, and an **objective function** (e.g., maximum of profit, performance, or yield)
 - Use optimization techniques to find the optimal solution to the objective function
 - “*The Science of Better*”
- OR has subtle differences from Industrial Engineering, Decision Analytics, Management Science, Business Analytics, Operations Management...

Main Sub-disciplines of OR

- Financial engineering
- Revenue management
- Manufacturing, service, and supply chain management
- Transportation
- ...

Example: Price American Options

- American option is one of the most trading options in financial markets
- Suppose you hold an American option for **Apple Inc.**, which will expire in **30 days**
- On each day, you can choose either “**Stop**” or “**Do nothing**”
 - If you stop, you will receive the stock price of Apple Inc. as the payoff and the entire decision-making process terminates
 - If you do nothing, you will receive 0 payoff
 - The stock price of Apple Inc. changes every day

Apple Inc + Follow

AAPL:US

Nasdaq GS (USD) · Market closed

208.37 ▲3.77 +1.84%

As of 8:00 PM EDT 04/24/25. Nasdaq Last Sale

Summary Related News Company Info Financials



Example: Price American Options

- The following questions may be helpful to build a mathematical model:
 - What is your objective in this example?
 - What is your decision?
 - What is the difficulty of making a good decision?
- Showcase of a (simplified) mathematical model

$$V(S_0) = \max_{\text{stopping day}} \mathbb{E}[\text{payoff when we stop}]$$

- or

$$V(S_0) = \max_{\tau \in \{1, \dots, 30\}} \gamma^\tau \mathbb{E}[h(S_\tau) \mid S_0 = s_0]$$

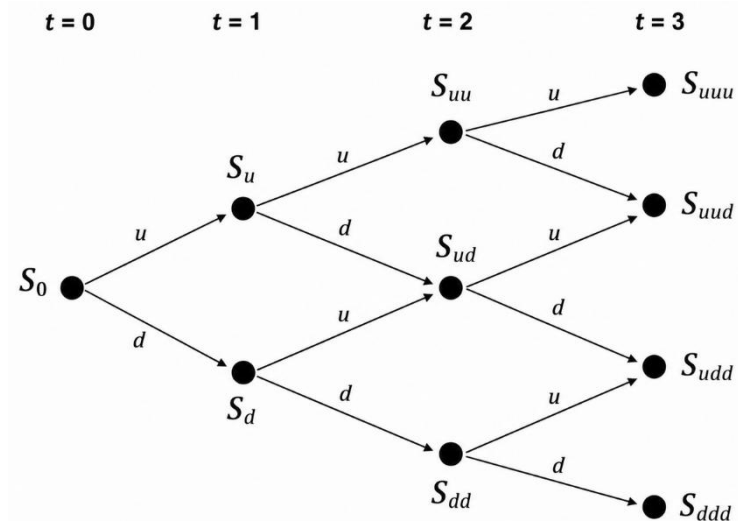
where $\tau \in \{1, \dots, 30\}$, γ is the discount factor, and the expectation is taken w.r.t. S_τ

Example: Price American Options

- Real problems are much more complex than our example
 - Obtaining an exact optimal solution **in closed form** is difficult in general
 - Multiple methods are used to solve the above model
 - Binomial trees, stochastic differential equations, deep neural networks, simulations...
 - **Near optimal solutions** can be obtained (e.g., errors within 5%)
 - Still an active research area: **1%** improvement on the payoff is considered **significant**

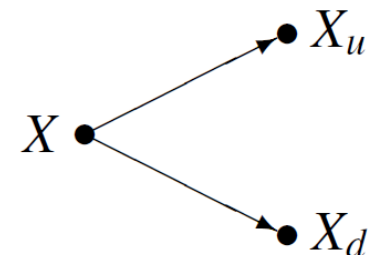
- Binomial tree

- Probability for a UP move: π_u
- Probability for a DOWN move: π_d



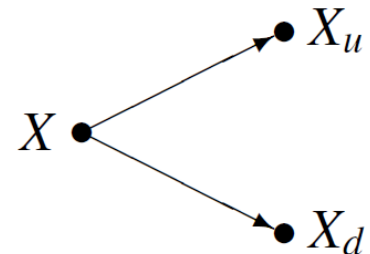
Example: Hedge risks in cash flows

- In one month, an energy trading company A can produce and sell on the Henry Hub, LA, wholesale market **10,000 units** of natural gas incurring a marginal cost of **\$3.5/Unit**
- Suppose the current price is **\$3.622/Unit** and we know there are two possibilities of the selling price in the considered month
 - With a probability **0.5691**, the price will go up to **\$3.8031/Unit**
 - With a probability **0.4309**, the price will go down to **\$3.2598/Unit**
- What is the expected (average) cash flow for the company?
 - If the price goes up, the cash flow is **$(\$3.8031 - \$3.5) * 10,000 = \$3031$**
 - If the price goes down, the cash flow is **\$0**
 - Thus, the expected cash flow is **$\$3031 * 0.5691 + \$0 * 0.4309 = \$1724.94$**
- What is the risk in this example?

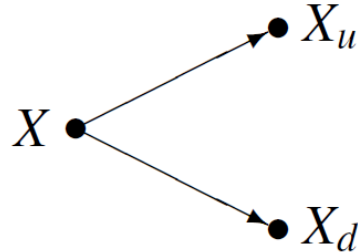


Example: Hedge risks in cash flows

- By **hedging**, we remove **randomness** from the cash flow
- Let F be the current Henry Hub natural gas **futures** price for next (prompt) month delivery; e.g., **$F = \$3.569/\text{Unit}$**
 - If we sell (short) a futures contract today, there is no current cash flow. However, in one month, we must **sell** certain amount of natural gas at the price **$\$3.569/\text{Unit}$**
- Suppose we sell **5578.87 units** of natural gas through the futures contract, then
 - If the price goes up to $\$3.8031$, the cash flow of this financial trading is
 $(\$3.569 - \$3.8031) * 5578.87 = -\$1306.01$
 - If the price goes down to $\$3.2598$, the cash flow of this financial trading is
 $(\$3.569 - \$3.2598) * 5578.87 = \$1724.94$



Example: Hedge risks in cash flows



Physical Trading

- If the price is \$3.8031:
 - \$3031
- If the price is \$3.2598:
 - \$0
- The expected cash flow is
 $\$3031 \times 0.5691 + \$0 \times 0.4309 = \$1724.94$

Financial Trading

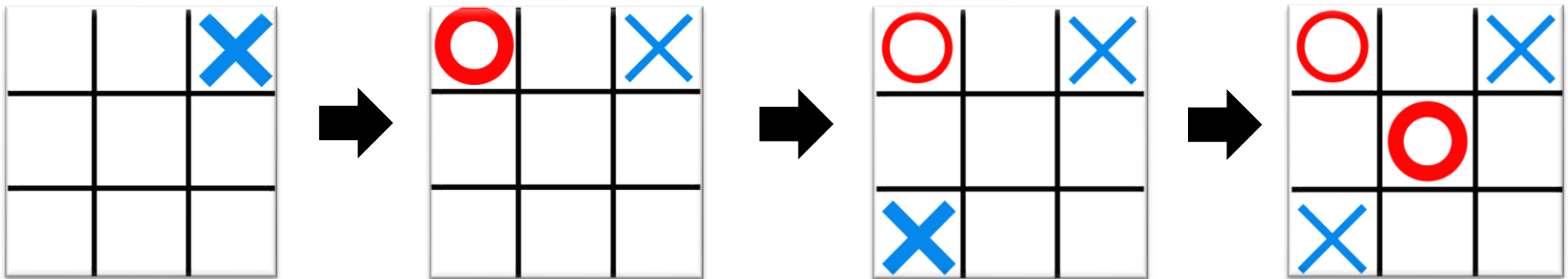
- If the price is \$3.8031:
 - -\$1306.01
- If the price is \$3.2598:
 - \$1724.99

Hedged Cash flow

- If the price is \$3.8031:
 - $\$3031 - \$1306.01 = \$1724.94$
- If the price is \$3.2598:
 - $\$0 + \$1724.99 = \$1724.94$
- The cash flow is exactly
\$1724.94 with probability one

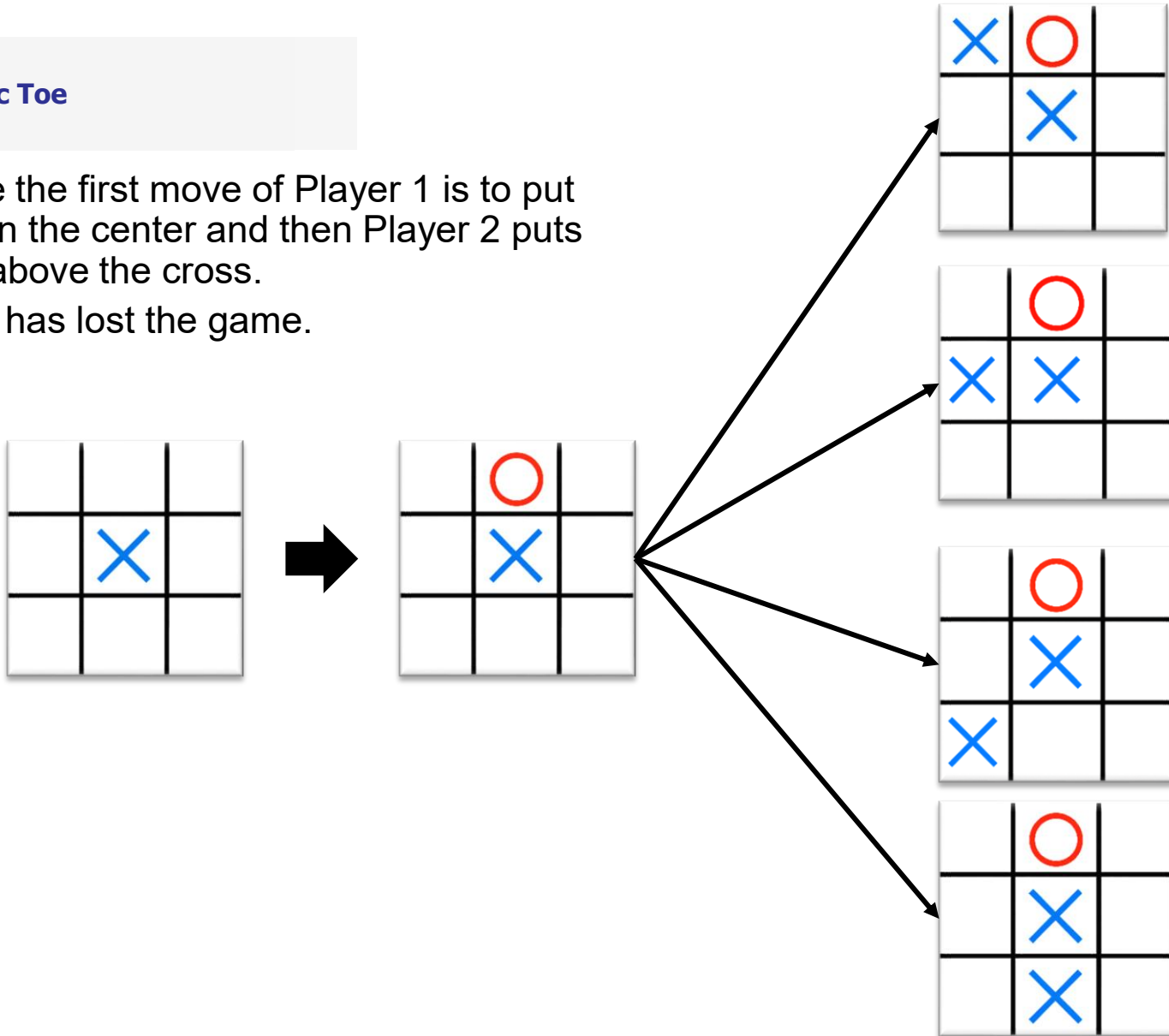
Example: Tic Tac Toe

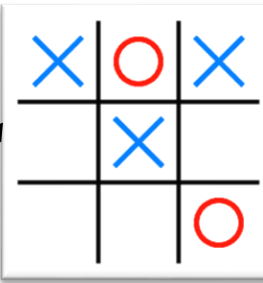
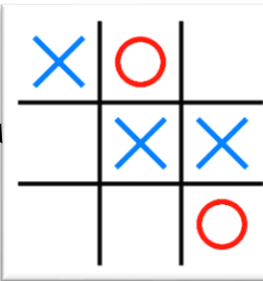
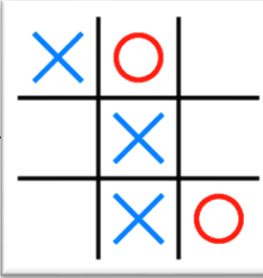
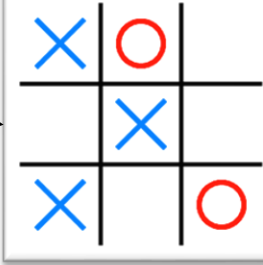
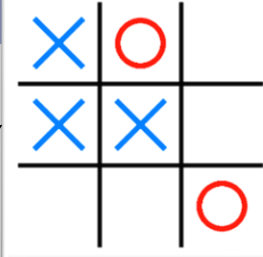
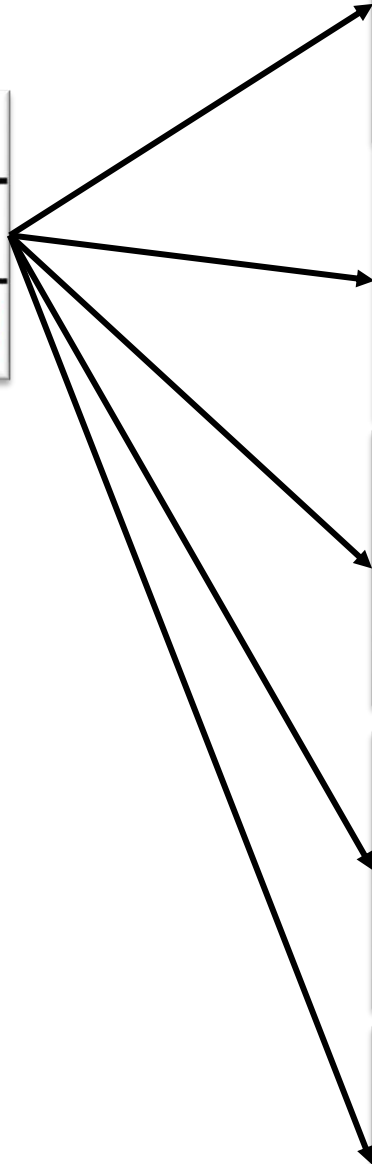
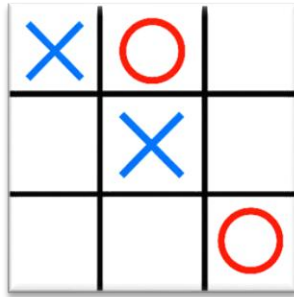
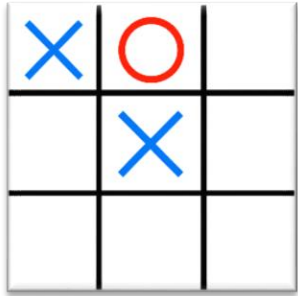
- A paper-and-pencil game for two players who take turns marking the spaces in a three-by-three grid, one with Xs and the other with Os.
- For example,



Example: Tic Tac Toe

- Suppose the first move of Player 1 is to put a cross in the center and then Player 2 puts a circle above the cross.
- Player 2 has lost the game.





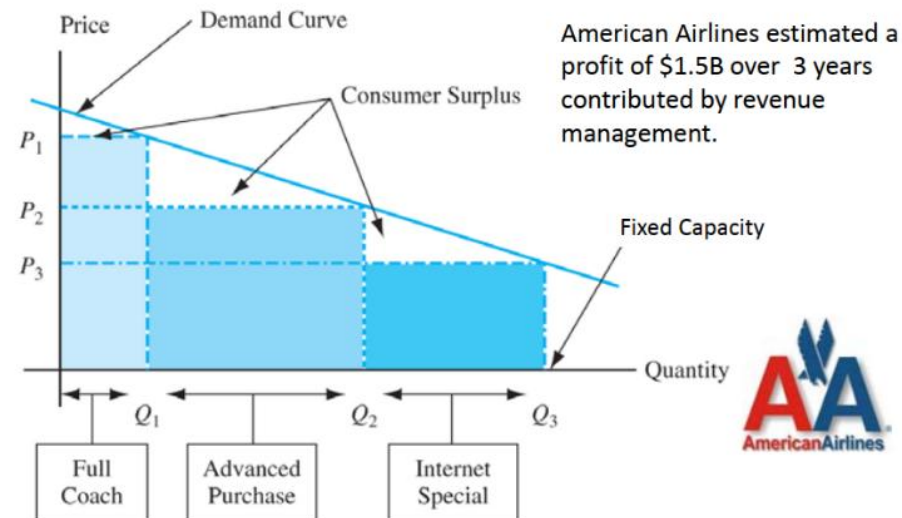
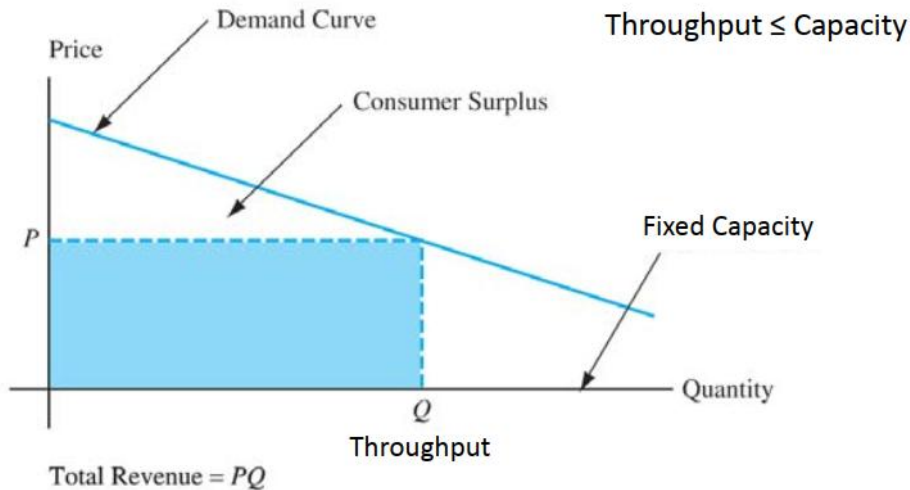
Example: Tickets Selling

- Suppose you're a manager of an airline. There is a flight from Hong Kong to New York city next month. There are **30 days left** for you to sell the tickets and there are **100 seats** on the flight. How would you set prices for these tickets?
- The following questions may help you to analyze the problem:
 - What's your objective?
 - Are you going to use the same price for every customer? Why?
 - Are you going to use a fixed price or change the price dynamically? Why?



Example: Tickets Selling

- The basic idea is to sell tickets to customers at different prices
- The difficulty is how to identify customers' true demand
 - Platforms tend to learn customers' behavior from their historical records



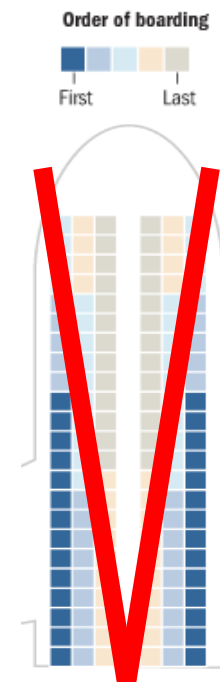
Example: Airplane Boarding

- Why is boarding strategy important?
 - Boarding time in 1960s: 20 passengers/minute
 - Nowadays: 9 passengers/minute as passengers bring along heavier carry-on luggage
 - Southwest: 10 minutes increase in boarding time needs 40 more planes at a cost \$40 million each to run the same number of flights
- What's the objective?
- What airplane boarding strategies have you seen in real life?



Example: Airplane Boarding

- US Airways introduced an innovative boarding strategy called “reverse pyramid” (2005)
- The strategy saves 26%~39% of the boarding time on average
- An average decrease of 21% in departure delays are observed



Example: Matching Supply and Demand

- **2001:** Microsoft launched Xbox, but it had to discount prices by over \$100 a year later due to 100,000 units excess supply
- **2005:** Microsoft launched Xbox 360, but 4.5M of units on backorder
- **2006:** Sony delayed launch of PlayStation 3 in Europe, Russia, and Australia until Mar. 2007, due to production delays
- **2006:** Nintendo lost \$1.3M global demand for its Wii video games console
- ...
- **2025:** Nintendo Switch 2 shortages could last until 2026



Tentative Schedule

Topic 1	Introduction
Topic 2	Linear programming
Topic 3	Integer programming
Topic 4	Probability Basics
Topic 5	Game Theory
Topic 6	Introduction to queuing theory
Topic 7	Introduction to dynamic programming
Topic 8	Decision making in investment

What we will focus on

- **Model**
- Algorithm Design
- Analysis
- **Implementation**
 - The course requires a lot of logical thinking and reasoning
 - The “pure” math part is limited

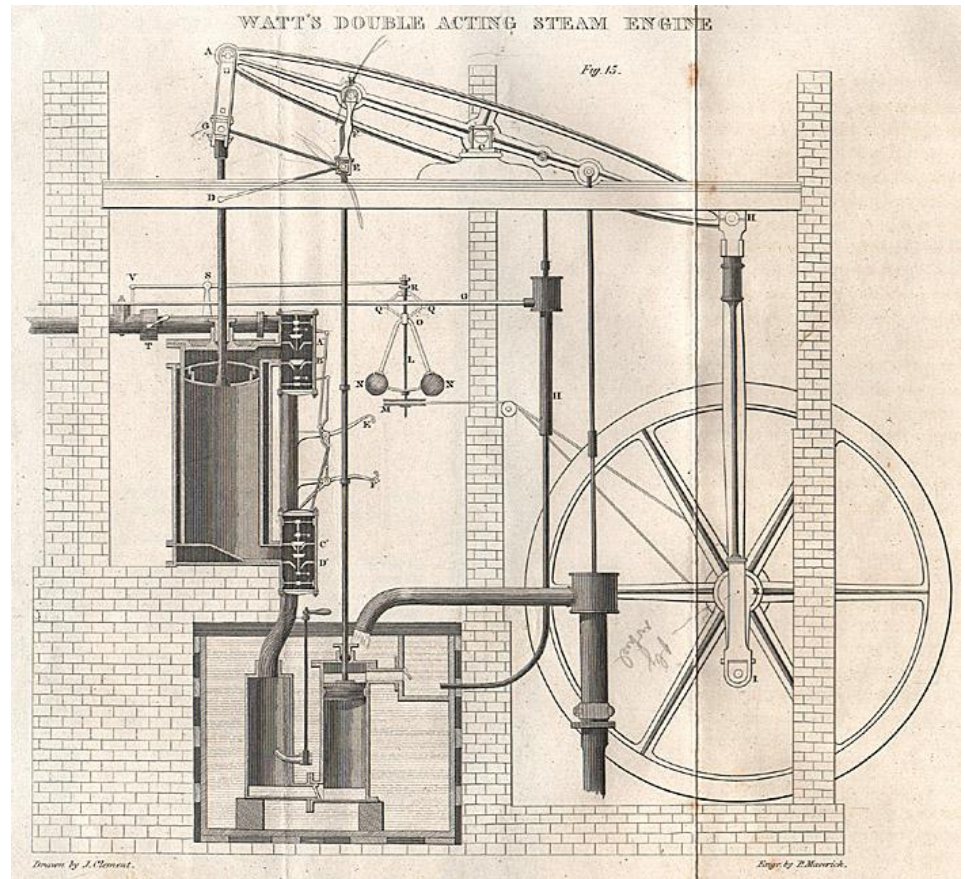
Thanks!

Early history of Optimization, Historical landmarks

1730-1800's: Industrial Revolution ***organized factories***



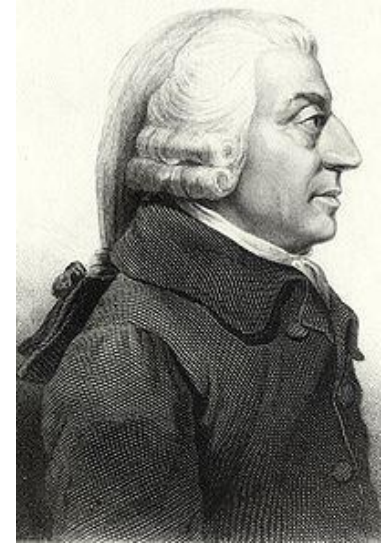
James Watt



Watt created a new and [much more efficient design](#) of the [Newcomen steam engine](#)

Division of Labor

- Adam Smith (1723-1790), The Wealth of Nations
 - Articulated the benefits of the division of labor
 - Explained the workings of the “invisible hand”
 - Proclaimed the beginning of modern capitalism
- Observed firms manufactured pins in two different ways:
 - Craft-style – each worker did all steps
 - Production – each worker specialized in one step



Division of Labor

■ Job specialization

- A division of labor occurs as different workers specialize in specific tasks over time
- Workers who specialized became much more skilled at their specific tasks
- Increasing job specialization increases efficiency and leads to higher organizational performance

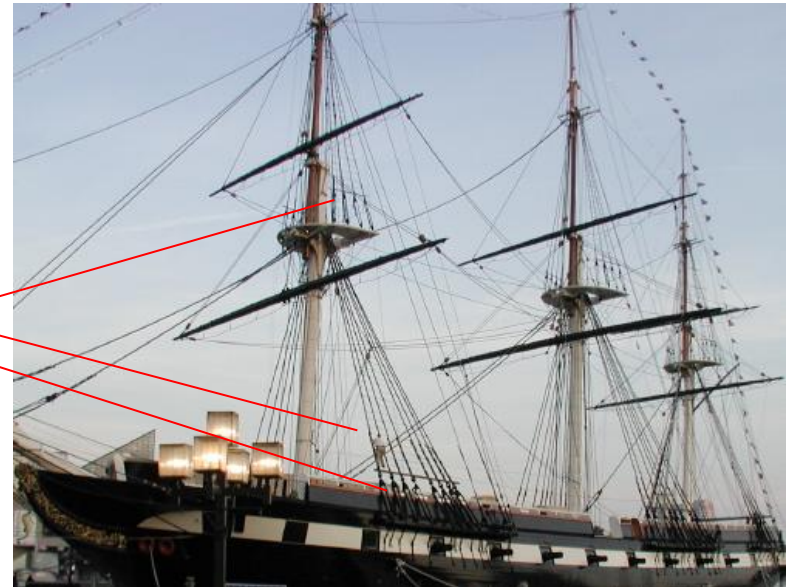
Interchangeability

- Interchangeable parts: identical parts (for practical purposes)
- Fit into any assembly of the same type, without any custom fitting (such as filing).
- Allows:
 - Easy assembly of new devices
 - Easier repair of existing devices
 - Minimizing both the time and skill required of the worker

Historical Landmarks....

1800s: Henry Maudslay

Interchangeability



- Interchangeable components for manufacture of rigging blocks
- Also introduced standard sized screws-nuts

Main requirement: Repeatable, Precision machining

Scientific Management

- Management is as old (older?) as the pyramids.
- Before early 20th century (maybe even now), management is often regarded as common sense, or art.
- Scientific management believes that
 - There is a science of management
 - Management can be studied
 - Managers can be trained

Scientific Management

- Frederick W. Taylor (1856 – 1915)

- Father of scientific management
- Father of industrial engineering



- Four principles of scientific management

- Rule-of-thumb work methods → methods based on scientific studies
- Scientifically select, train, and develop each worker rather than passively leaving them to train themselves.
- Cooperate with workers to ensure that the scientifically developed methods are being followed.
- Divide work nearly equally between managers and workers
 - Managers apply scientific management principles to planning the work
 - Workers perform the tasks

Scientific Management

- Planning vs. Doing
 - Managers plan (define tasks, set standards, ...)
 - Workers work



Mass Production

- Henry Ford (1863 – 1947) pushed division of labor and work standardization to perfection.
 - Innovation of the moving assembly line.
 - The assembly line is divided into small stations.
 - A worker repeats short cycles of identical work.
 - All feedlines along the route were synchronized to supply the right parts, at the right time.



Mass Production

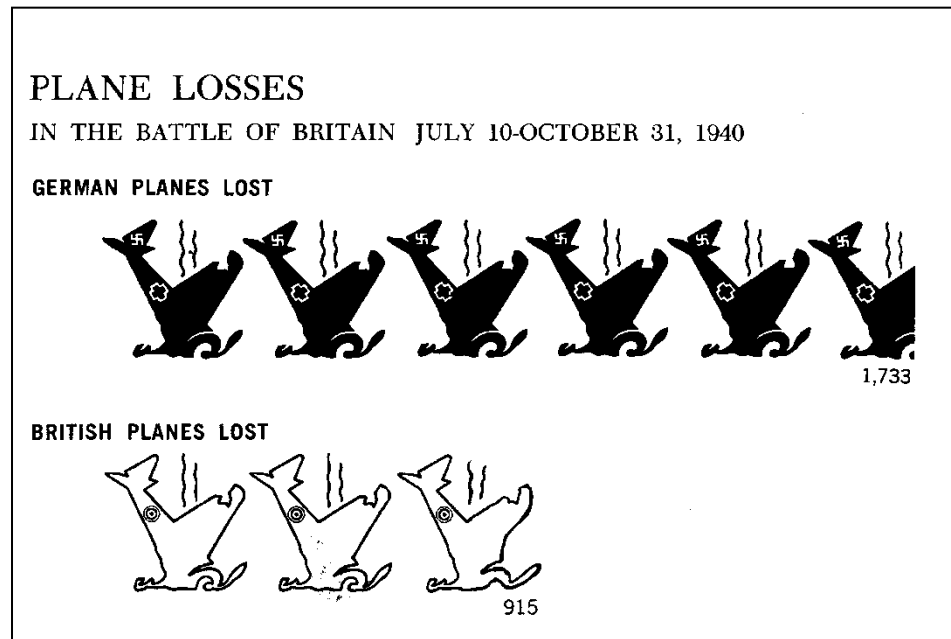
- This new manufacturing system produced cars very quickly and efficiently. So the price became much cheaper.

Year	Price of Model T
1908	\$850
1916	\$360
1920's	\$250

- Model T (1908-1927)' s total production was 15,007,034.
- By 1918, half of all cars in America were Model T.
- Model T was all black (for assembly convenience), as Ford wrote in his autobiography, “Any customer can have a car painted any color that he wants so long as it is black”.

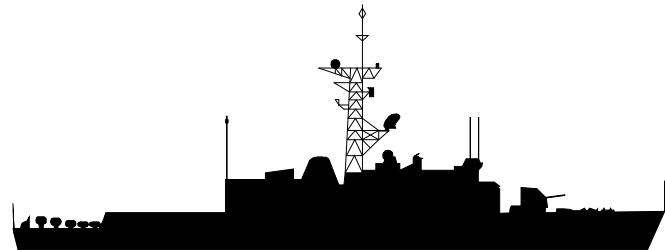
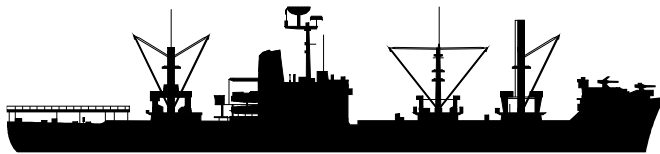
The Rise of Operations Research

- Operations Research (OR) is the field of how to form mathematical models of complex management decision problems and how to analyze the models to gain insight about possible solutions.
- OR started in World War II in U.K. for better air defense.



The Rise of Operations Research

- It was then adopted by the U.S. Navy. For example
 - **Problem:** Increase movement of war time supplies with limited shipbuilding capacity
 - **Question:** Build more merchant or more escort ships



- Each escort ship saves 2 to 3 merchant ships per year



The Rise of Operations Research

- After WWII, OR started to spread out of military use.
- Objective: Find the most effective utilization of **limited** resources by the use of quantitative techniques.
- Computing technique advance also significantly helped the widespread of OR.
- OR is sometimes called management science or decision science, and recently evolves into **business analytics / decision analytics** (using **big data**).
- It includes many quantitative techniques, including mathematical programming, stochastic process, queueing theory, game theory, etc.

Examples of Industrial Engineering Applications

- Toyota's Just-in-Time production system



- Zara's quick response supply chain

Z A R A

- Southwest's low-cost flights

